

README

File location: `$VCS_HOME/doc/examples/nativetestbench/openvera/rvm/restart/README`

This directory contains example on how to use the `rvm_env::restart()` method and how to structure tests to allow easy sequencing in the same simulation.

Note that this is not the default recommended approach in a random-based coverage driven approach. Tests should be short enough to be able to be run in sequence as individual simulations. However, certain DUT require a significant simulation time to be configured before the active portion of the test can be executed. It may be desirable, in regression tests, to be able to sequence a series of compatible tests to minimize the overhead associated with DUT configuration.

The first example (`hard`), shows the simplest approach and handles arbitrary tests but requires that the DUT be reset and reconfigured between tests (which is exactly what you may be trying to avoid by sequencing tests in a single simulation).

The second example (`soft_simple`), shows the simplest approach without resetting the DUT but does not guarantee random stability. That means that running test "01" after test "00" in the same simulation will yield a different random stimulus sequence (and therefore a different test) than running test "01" on its own or first in the sequence.

The last example, (`soft_stable`), shows the more complex approach but satisfies random stability requirements. Tests can be run in any order and will yield the same stimulus regardless of their position in the sequence or run individually. This is achieved by the careful implementation of the `rvm_env::save_rng_state()`, `rvm_env::restore_rng_state()`, `rvm_xactor::save_rng_state()` and `rvm_env::restore_rng_state()` methods.

The examples for soft restarts (`soft_*`) require a sequence of tests to use the same configuration. A new DUT configuration is not generated nor uploaded nor is the DUT reset between tests. Such tests cannot call the `rvm_env::gen_cfg()`, `rvm_env::build()` and `rvm_env::cfg_dut_t()` methods explicitly.